

Week 4: Web Application Development (Yo SACCO)

We shall develop a mini-core banking system for a Savings and Credit Co-operatives (SACCO) group called **Yo SACCO**. Savings and Credit Co-operatives (SACCOs) are community membership-based financial institutions, that are formed and owned by their members in promotion of their economic interests. They save money and lend it among themselves at an agreed interest rate. The revenue generated out of the loan interest when members repay their loans helps to run the SACCO activities.

Yo SACCO Functions to SACCO Member Admins

The Yo SACCO system shall provide the following functions to the SACCO members.

1. Create, update and delete members. Each member profile should consist the following attributes.
 - id (Integer)
 - Name (50 String)
 - Gender (Male or Female) (10 String)
 - Date of Birth (Date)
 - Account Number (50 String)
 - Phone: A validate member's phone number in the format (CCCNXXNNNNNN e.g. 256772123456) (50 String)
 - PIN (Should be hashed using SHA256 algorithm) (255 String)
 - Next of Kin Name (255 String)
 - Create date: This is the date when the member's account was created (Date).
 - Role (Chairperson, Secretary or Treasurer) (50 String)
 - Available Balance: This column should store the member's available balance. (double)
2. Record member cash deposits. Each deposit should have the following attributes:
 1. Member ID (integer)
 2. Amount (double)
 3. Description (255 String)
 4. Payment method. Each Payment method should have the following attributes:
 1. id (integer)
 2. Name (50 String)
 3. Type (Cash, Bank Transfer, Mobile Money) (50 String)
 5. Date and Time (Datetime)
3. Record member cash withdraws. Each withdraw transaction should have the following attributes:
 0. Member ID (integer)
 1. Amount (double)
 2. Description (255 String)
 3. Payment method. Each Payment method should have the following attributes:
 1. ID (integer)
 2. Name (50 String)

3. Type (Cash, Bank Transfer, Mobile Money) (50 String)
4. Date and Time
4. Generate member statement: The system should have a section where an authorized member may generate a member's statement. Each statement record should have the following attributes:
 0. Create date & time (Datetime)
 1. Member Name (50 String)
 2. Transaction Type: This can be set to (DEPOSIT, WITHDRAW, LOAN REPAYMENT, or any other transaction type) (50 String)
 3. Payment method: This can be set to (CASH, MOBILE MONEY or BANK TRANSFER) (50 String)
 4. Description (255 String)
 5. Member Balance (double)
5. Member electronic deposit: This function shall allow an individual member to initiate a deposit using mobile money. Yo! Payments shall handle the processing of mobile money.
6. Member electronic withdraw: This function shall allow an individual member to initiate a withdraw which should go through an approval process. The approvals shall be done by the SACCO executive members (that's the chairperson, secretary and the treasurer).
7. Create loans: To be defined in future.

Yo SACCO Functions to Yo Backend Admin

The following are the functions the application shall provide to the Yo Uganda staff in charge of managing SACCOs.

1. Create, update and delete SACCOs. Each SACCO should have the following attributes:
 1. id (integer)
 2. Group ID (it should be a 4-digit string and unique)
 3. Name (50 String)
 4. Location (50 String)
 5. Created On (Datetime)
 6. Members: A single SACCO may have one or more members. Each member should have all the attributes mentioned in **Yo SACCO Functions to SACCO Administrators** section above.
2. Manage Access Control Groups (ACG)s: This shall include create, update and delete ACGs. of Access Control Groups are grouped privileges that can be assigned to a single user account or group of user accounts. A privilege is definition of an action that can be performed by a user on the system. Each privilege should have the attributes including (Id, Name). A list of pre-defined attributes shall include the following:
 1. View Users
 2. Create Users
 3. Update Users
 4. View SACCOs
 5. Create SACCOs
 6. Update SACCOs
 7. Delete SACCOs

8. View ACGs
9. Create ACGs
10. Update ACGs
11. Delete ACGs
3. Each ACG shall have the following attributes:
 1. id (integer)
 2. name (50 String)
 3. privileges (This should be an array of privileges).
4. Create, update and delete users: Users are Yo staffs who may login to the Yo SACCOs system to create SACCOs, view SACCO transactions, manage other users, etc. Each user account should comprise of the following attributes:
 1. id (integer)
 2. Name (255 String)
 3. Phone number (20 String)
 4. Email address (unique)
 5. Create date (Datetime)
 6. Last Updated (Timestamp)
 7. AGCs: Each user may be assigned one or more AGCs accounts.

Stack

The mini-core banking system shall be composed of the following:

1. SACCO Members' web portal.
2. Backend processing (written in Python using Flask Framework). Learn this framework from here (<https://flask.palletsprojects.com/en/3.0.x/>). Particularly, you will need to learn how to use Authentication, Templates, Routing, Session and Database Connection.
3. Database Component. The preferred choice is MySQL and the schema has been defined below. You will have to learn how to write SQL queries (including INSERT, UPDATE AND SELECT). You can use the following reference for this: <https://www.w3schools.com/MySQL/default.asp>
4. Backroom Admin portal.

Database Schema

Your records shall be stored in a MySQL database named yosacco whose schema is defined below:

Database Schema

```
CREATE TABLE `user` (  
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(255) NOT NULL DEFAULT '',  
    phone VARCHAR(255) NOT NULL DEFAULT '',  
    email VARCHAR(255) NOT NULL DEFAULT '',  
    password VARCHAR(255) NOT NULL DEFAULT '',  
    created_on DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    updated_on TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE  
CURRENT_TIMESTAMP,  
    CONSTRAINT `UK_user_unique_email` UNIQUE KEY `UK_user_unique_email`  
(`email`)  
) Engine = InnoDB;  
  
CREATE TABLE `acg` (  
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(255) NOT NULL DEFAULT '',  
    created_on DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    updated_on TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE  
CURRENT_TIMESTAMP,  
    CONSTRAINT `UK_acg_unique_name` UNIQUE KEY `UK_acg_unique_name`  
(`name`)  
) Engine = InnoDB;  
  
CREATE TABLE `user_acg` (  
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
    user_id BIGINT UNSIGNED,  
    acg_id BIGINT UNSIGNED,  
    created_on DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    updated_on TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE  
CURRENT_TIMESTAMP,  
    CONSTRAINT `UK_user_unique_acg` UNIQUE KEY `UK_user_unique_acg`  
(`user_id`,`acg_id`),  
    CONSTRAINT `Fk_user_acg_user_id` FOREIGN KEY `Fk_user_acg_user_id`  
(`user_id`) REFERENCES `user`(`id`) ON DELETE CASCADE,  
    CONSTRAINT `Fk_user_acg_acg_id` FOREIGN KEY `Fk_user_acg_acg_id`  
(`acg_id`) REFERENCES `acg`(`id`) ON DELETE CASCADE  
) Engine = InnoDB;  
  
CREATE TABLE audit_trail (  
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
    user_name VARCHAR(255) NOT NULL DEFAULT '',  
    email VARCHAR(255) NOT NULL DEFAULT '',  
    action VARCHAR(255) NOT NULL DEFAULT '',  
    data TEXT,  
    created_on DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    updated_on TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE  
CURRENT_TIMESTAMP  
) Engine=InnoDB;
```

```

CREATE TABLE `sacco` (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `group_id` VARCHAR(255) NOT NULL DEFAULT '',
  `name` VARCHAR(255) NOT NULL DEFAULT '',
  `location` VARCHAR(255) NOT NULL DEFAULT '',
  `created_on` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  `updated_on` TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
  CONSTRAINT `UK_sacco_group_id` UNIQUE KEY `UK_sacco_group_id`
(`group_id`)
) Engine = InnoDB;

CREATE TABLE `sacco_member` (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `sacco_id` BIGINT UNSIGNED,
  `account_number` VARCHAR(255) NOT NULL DEFAULT '',
  `pin` VARCHAR(255) NOT NULL DEFAULT '',
  `fname` VARCHAR(255) NOT NULL DEFAULT '',
  `lname` VARCHAR(255) NOT NULL DEFAULT '',
  `gender` VARCHAR(10) NOT NULL DEFAULT '',
  `phone` VARCHAR(255) NOT NULL DEFAULT '',
  `email` VARCHAR(255) NOT NULL DEFAULT '',
  `role` ENUM('CHAIRPERSON','TREASURER','SECRETARY','REGULAR MEMBER')
NOT NULL DEFAULT 'REGULAR MEMBER',
  `balance` DOUBLE DEFAULT 0,
  `next_of_kin_name` VARCHAR(255) NOT NULL DEFAULT '',
  `date_of_birth` DATE,
  `created_on` DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  `updated_on` TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
  CONSTRAINT `UK_sacco_unique_member_id` UNIQUE KEY
`UK_sacco_unique_member_id` (`sacco_id`,`account_number`)
) Engine = InnoDB;

CREATE TABLE `transaction_log` (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `sacco_id` BIGINT UNSIGNED NULL,
  member_id BIGINT UNSIGNED NULL,
  amount DOUBLE NOT NULL DEFAULT 0,
  narrative VARCHAR(255) NOT NULL DEFAULT '',
  reference VARCHAR(255) NOT NULL DEFAULT '',
  status VARCHAR(255) NOT NULL DEFAULT '',
  transaction_type ENUM('DEPOSIT','WITHDRAW','LOAN REPAYMENT','NONE')
NOT NULL DEFAULT 'NONE',
  payment_method ENUM('CASH','BANK TRANSFER','MOBILE MONEY','NONE') NOT
NULL DEFAULT 'NONE',
  `trace` TEXT NULL,
  `check_status_trace` TEXT NULL,
  balance_before DOUBLE NOT NULL DEFAULT 0,
  balance_after DOUBLE NOT NULL DEFAULT 0,
  created_on DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
  updated_on TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
  completion_date DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT `UK_transaction_log_reference` UNIQUE KEY
`UK_transaction_log_reference` (`reference`),

```

```
        CONSTRAINT `FK_UK_transaction_log_member_id` FOREIGN KEY
`FK_UK_transaction_log_member_id
`(`member_id`) REFERENCES `sacco_member`(`id`),
        CONSTRAINT `FK_UK_transaction_log_sacco_id` FOREIGN KEY
`FK_UK_transaction_log_sacco_id`(`sacco_id`) REFERENCES `sacco`(`id`)
ON DELETE SET NULL
) Engine=InnoDB;
```

About Frontend Interface Design

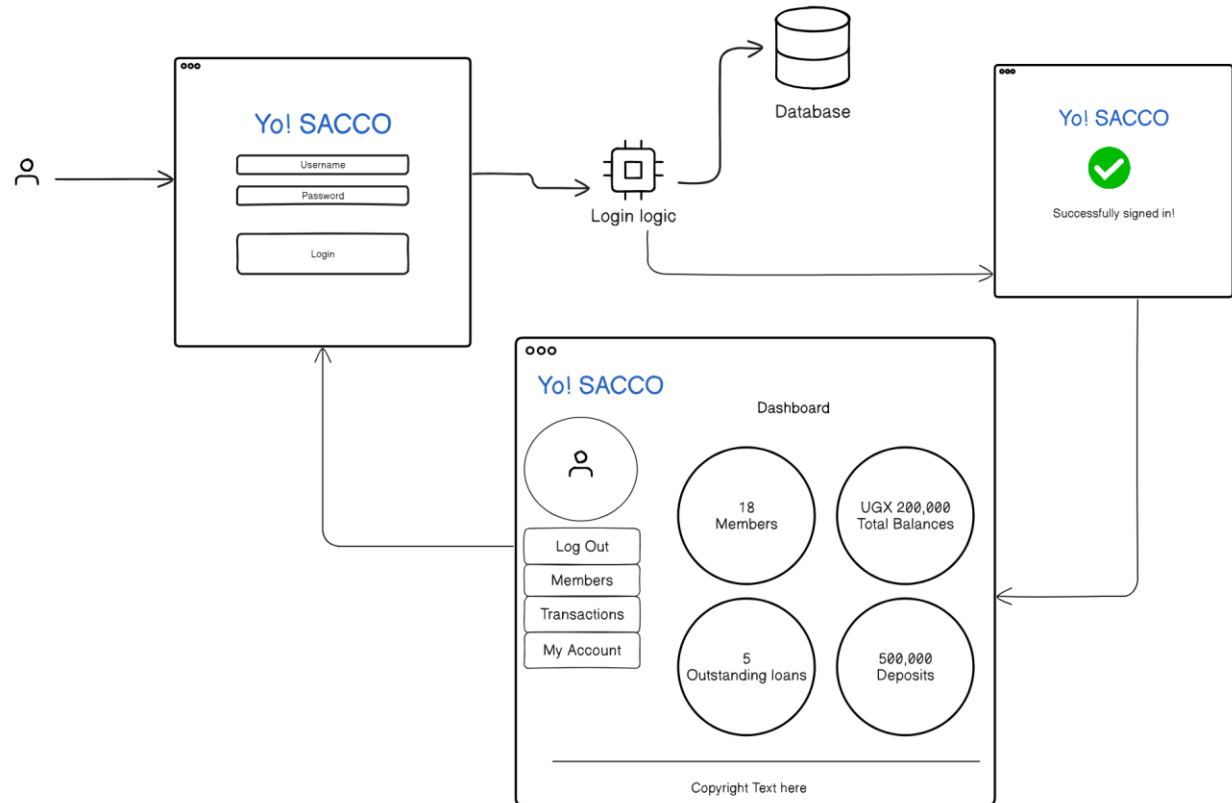
The frontend interfaces shall be developed in HTML, CSS and JavaScript. Learn how you can build frontend User Interfaces using JQuery framework called EasyUi from here

<https://www.jeasyui.com/index.php>. Regarding our project here, you need to specifically learn the following:

1. Layouts: You will need this to organize your user interface elements.
2. menus: You will need this to create a navigation list from which the user can move to various pages or contents.
3. Forms: You need this to create forms such as the customer creation form, deposit form and withdraw form
4. Grid. You will display your tabular data using the grid component.

Login Process

The following is an illustration of the login process:



Some of the Application View Section

General Layout

Yo! SACCO

Log Out

Members

Transactions

My Account

Members

Add

Search Member

ID	Name	Phone	Balance	Action
1	James K	256772	200,000	edit view
2	Ruth	256773	500,000	edit view
3	Juma	256774	400,000	edit view
4	Jane	256775	150,000	edit view
5	Richard	256775	300,00	edit view

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Transactions

Yo! SACCO

Log Out

Members

Transactions

My Account

Transactions

Search Member

ID	Datetime	Member	Type	Amount
1	12/09/2024	James	DEPOSIT	20,000
2	12/09/2024	James	DEPOSIT	50,000
3	12/09/2024	Richard	WITHDRAW	400,000
4	12/09/2024	Jane	WITHDRAW	300,000
5	12/09/2024	Jane	DEPOSIT	10,000

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My Account

Yo! SACCO

Log Out

Members

Transactions

My Account

Statement

Print

BAL: 1,000,000

ID	Datetime	Type	Amount	Balance
1	12/09/2024	DEPOSIT	+200,000	10,00,000
2	12/09/2024	DEPOSIT	+500,000	800,000
3	12/09/2024	WITHDRAW	-50,000	300,000
4	12/09/2024	WITHDRAW	-50,000	350,000
5	12/09/2024	DEPOSIT	-10,000	400,000

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Member Creation Form

Yo! SACCO

Log Out

Members

Transactions

My Account

New Member

First Name:

Phone:

Last Name:

Location:

Gender:

Male

Role

Chairperson

Date of Birth:

Submit

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Backend Admin Screens

Yo Backend

Yo! SACCO



Log Out

SACCOs

Transactions

Audit Trail

Users

ACGs

Users

Add

Search User

ID	Name	Phone	Email	Action
1	James K	256772	mm@domain.io	edit view
2	Ruth	256773	ruth@gmail.com	edit view
3	Juma	256774	juma@outlook.com	edit view
4	Jane	256775		edit view
5	Richard	256775	richard@mail.com	edit view

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SACCOs Section

Yo! SACCO



Log Out

SACCOs

Transactions

Audit Trail

Users

ACGs

SACCOs

Add

Search SACCO

ID	SACCO ID	Name	Members	Action
1	1001	Twezimbe	30	edit view
2	1002	Togetherness	20	edit view
3	1003	Collective	15	edit view
4	1004	Fenna	40	edit view
5	1005	GMasaka	60	edit view

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